Ocular Anatomy Review

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Ocular Adnexa

- Be able to identify the external features of the eyelids
- Describe the functions of each structures that make up the eyelids
- Describe the function of each of the muscles of the eyelids
- Describe the difference of ptosis appearance depending on the muscle involved
- Describe how the glands of the eyelid contribute to the different tear layers
- Describe how the tears drain from the orbit

2 ablauf von tränendrüse bis nase

picture

tarsal plate
~~orbital plate~~

levator muller helps keep eye up

aquaous and acessory

goblet cells, muzin

Cornea, Conjunctiva and Sclera

- Describe the average central corneal thickness
- Describe the clinical significance of the five corneal layers
- Describe the layers and structures involved with a recurrent corneal erosion dystrophy
- Describe where the cornea obtains oxygen and nutrients tears,
- Describe the functions of the sclera, episclera, Tenon capsule, and conjunctiva

3 klare anordnung = klare Sicht wichtig

535-555 im zentrum

epithelium, protect water coming in for a clear cornea

2 und 3 resistent to damage, 3 is very regular fibers

descemet, basement membrane of endothelium
termination of descemet (schwalbe Linie)

hemidesmosom problem in dystrophy problem

functin sclera forms eyeball
episclera drainage blood supply, sending vessels to c
tenons capsle protect globe, tough tissue
conjb. smooth eyemovement

Lens

- Describe how the embryonic nucleus forms
- Describe how the lens fibers form around the embryonic nucleus
- Describe how the Y sutures can be used to determine in which nucleus a cataract is located
- Identify the lens capsule, Y sutures, and embryonic nucleus

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posterior epithelium growth forward and goes in this space
embryonic nucleus

fibre start of equadot and goes iin the mittle and forms sutures

part 3: anterior y posterior is inverted, if its between

embryonic nucles,
opacity along after suture --> phetal

just have a look of the picture what the parts look like

iris cb choroid

colarettes, iris roots (thinnest area) (peripheral termination of iris (gets wrapp

Uvea

- Identify and describe and the landmarks present on the iris
- Name the iris and ciliary blood vessels
- Discuss how the iris and ciliary blood vessels and the ciliary body epithelium contribute to the blood aqueous barrier and uveitis
- Discuss the functions and innervation of the iris and ciliary body muscles

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major circle of the iris found in ciliary body

radial blood vessels gehen von aussen nach innen

minor blood vessels

iris blood vessels are non fenestrated

ciliary body is fenestrated (tight junctions)

supply sphincter and müller's : parasyhmpathic

sympathic dilator

Uvea (continued)

- Name and recognize the location and characteristics of the two main parts of the ciliary body
- Discuss the vascular patterns of the arteries of the choroid and choriocapillaris
- Describe the location of vortex veins draining blood from the choroid
- Discuss how Bruch's membrane is involved in age-related macular degeneration

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pars plana (more posterior) and pars plicata

bloodvessels get smaller and smaller, go to

single layer blood vessels in RPE

in quadrants, mind. 4 vortex veins, up to 10-12

drusen, phospholipid build up in bruchs membran

fat doesn't like water

no water in to the retina and no waste products
Out of retina --> starting in edema

drusen = phospholid

IOP and Aqueous Chambers

- Compare the three chambers of the globe
- Identify the anatomical appearance of the anterior chamber angle structures with gonioscopy
- Describe ways that aqueous exits the anterior chamber

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anterior chamber, posterior chambre, vitreous chamber

CB, skleral sporn, trabecel meshwork, schwalbe Linie

the flow of cunventional and unconventional

--> will be a flow chart!

RPE, 5 mainn Purposes,limiting of nutrition and waste, phagozytose

vitamin A, productes VEGF, absorbts light

Retina

- Describe the functions of the RPE
- Contrast the types of photoreceptors and under what conditions they are stimulated
- Describe the portions of and functions the photoreceptor
- Identify the types, functions, and synapses of ganglion cells

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rods and cones

rods stimulated dim, cones in bright

segment, fibre, outer and inner segment

outer and inner fibre

P (parvo) and M ganglion cells, intrinsicly photosensitive ganglion cells

P = Detail, contrast, M moving

synapse: ganglion synapse 4 places in brain
Lateral GN, pretectal nucleus, superior colliculus, hypothalamus
--> hypothalamus for circadian rhythm

Retina (continued)

- Identify each of the ten retinal layers on an OCT
- Identify the photoreceptor parts on an OCT
- Describe the structures and/or connections found in each of the 10 retinal layers
- Describe the position of the macula relative to the disc

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picture OCT,

OCT: 3 bright lines at botom, first RPE, Outer segment (dis

inner segment (elipsoid)

dim line: external limited membrane

inner nucleur: bipolar, horizontal, amakrine, muller cells and blood vessels

nerve fibre = Axons of Ganglien cells

inner limital membrane = müllers cell!!!!

position of macula little inferior and temporal 17°

Retina (continued)

- Describe the anatomic differences between the fovea and the remainder of the retina, including the relative concentration of photoreceptors and blood vessels
- Describe the location, patterns, and structures supplied by the retinal and choroidal blood vessels
- Describe how the anatomy contributes to the blood retinal barrier

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layers: RPE, outer segment, inner segment
~~external lim membran, inner limited membrana~~
outer nucleur, no outerplex
~~nicht alles,~~

no rods in centrum, rod free zone, no capillary noblood ves
~~extra cones in centrum~~

inner nucleur layer --> inner retina

blood vessels in nerve --> inner retina

choroid für photoreceptors

choroidvessels are fenestrated

RPE has tight junctions, zonula occludens

Orbit

- Draw and label the bones that contribute to each of the orbital walls
- Describe the landmarks associated with each orbital bone, as well as the structures that are associated with each landmark
- List the nerves and blood vessels that travel within each of the canals, foramen, and fissures of the orbit

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pyramide zeichnen können

what bone lacrimal glande?

supraorbital nodge supraorbital vessels

supratrochlea: supratrochleary

infraorbital foramen and canal and groove:
~~infraorbital nerve (lower eyelid)~~

inferior orbital fissure: same as ?

superior orbital fissure (nerves: C3, 4,5 nur V1 und V2, 6) und sympathatic fibres

optic canal: optic nerve, central ophthtlamic artery

Circulation

- List the branches of the ophthalmic artery and the ocular structures to which each branch supplies blood.
- List the branches of the external carotid artery and the ocular structures to which each branch supplies blood.
- Describe the blood vessels that supply the anterior and posterior choroid (outer retina) and differentiate this from the blood vessels that supply the inner retina.

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drying the flowsheet and add each of those bv (bild)

choroid photoreceptors, retina supply the inner
~~by retinal blood vessels~~

Circulation

- Describe how blood drains from the choroid and contrast this with how blood drains from the retina.
- Describe how the palpebral arcades form and where they are located within the eyelids
- Describe the blood vessels that supply the optic nerve.
- Describe the anastomosis that occurs to form the major circle of the iris.

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choroid veins: vortexveins, Retina central retina vein

palpebral arcade, lacrimal artery turns into the lateral palpebral arcal

short posterior ciliary artery a bit central retinal artery

posterior long artery, anterior ciliary artery

Extraocular Muscles

- Describe the location of the origin and insertion of each of the extraocular muscles.
- List what nerves innervate each of the extraocular muscles
- List the primary and secondary actions of each extraocular muscle.

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commn tendon ring (annulus of zinn)
trochlea
SO: trochlea, insertion for ex
IO: posterior lateral??

CN3, MR, IR, SR, IO
CN4 SO
CN6 LR

SR 1. elevation, intorsion adduction
SO prim intorsion, depression and abduction
IR: 1. depression, extorsion, abduction

Extraocular Muscles (cont.)

- Describe muscle movements as the eye moves from a secondary position to a tertiary position, and use this to determine dysfunctional muscle.
- Determine which extraocular muscles are agonist and antagonists when moving about each of Fick's axes.
- Determine the yoke muscles involved in each secondary and tertiary eye movement

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which one is disfunctional? X, Y, Z

SR, SO agaonist in torsional movement,
antagonist in up and down, antagonist in lateral??

yoke muscle mean both eyes are working together
one eye is abducted, one eye is adducted

nach oben links: R IO, L SR are yoke muscles

Cranial Nerves

- List the branches, pathways, and functions of each branch of the ophthalmic nerve.
- List the branches, pathways, and functions of the branches of the maxillary nerve that innervate structures surrounding the orbit

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nasociliary nerves, frontal 2 branches, lacrimal 0 branches

put things innervate

maximallry: infraorbital nerve

Cranial Nerves (cont.)

- Describe the central, preganglionic, and postganglionic sympathetic pathways to the eye and ocular structures.
- Describe how the sympathetic system affects the eye and ocular structures.
- Describe the central, preganglionic, and postganglionic parasympathetic pathways to the eye or ocular structures.
- Describe the parasympathetic system affects the eye and ocular structures.

more accommodation and miosis

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central: brain stamm, superior ganglion etc.

post. internal carotry cavernosus sinus dilatar and muller

long ciliary, dilataror, muller cranial n3

pupilles sympathic: dialate, muller enlarge eye

hypothalamus edinger wes. cranial 3 to ciliary ganglion

then short ciliary nerves

Cranial Nerves (cont.)

- Describe the pathways of cranial nerves 3, 4, and 6.
- Describe the characteristic of a cranial nerve 3, 4, or 6 palsy.
- List the functions and pathway of the branches of cranial nerve 7 that innervate muscles surrounding the eye.

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4 cavernosus sinus

6 is in pons into the sinus cavernosus

CN7 lagopthtlamus

Pupils

- Describe the afferent, central, and efferent structures involved in the pupil reflex
- Describe the characteristics of a relative afferent pupillary defect and how the pupils would respond when there is an afferent pupillary defect.
- List the causes of a relative afferent pupillary defect and describe how these conditions would present clinically

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aff = retina optic nerve chiasm, optic tract superior colliculus

central betweeen pretectal and EW efferent vorne

RAPD no anisocoria

very large retinal lesion of RAPD, small optic nerve lesion cause RAPD

optic tract: contralateral RAPD, supra colliculus contra visual field to test, Optic tract homonomy hemionopia no visual issues when supr coll?

Pupils (continued)

- Describe the pathway for the near pupillary response and how that differs from the light pupillary pathway.
- Describe the pupil characteristics when there is a lesion in the area of the central pupillary fibers.

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near pathway pretectous ... siehe bild, light near dessociation

light near dessociation

Pupils (continued)

- Describe the pupillary responses and general characteristics when there is a lesion affecting any part of the parasympathetic pupillary pathway
- Describe the pupillary responses and general characteristics when there is a lesion affecting the sympathetic pupillary pathway

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parasympathic: auge dilated, worse in bright illumination

ptosis CN3, segment constritction is tonic pupille (cruise holiday?

sympathtetic: horner: trial: Ptosis miosis anhydrosis

small pupill, ptosis (little) muller, anhydrosis (lag of swetting)

dylatation lag

5 sec after, 15 after siehe Dokumnetation

Pupils (continued)

- Describe the pharmacologic testing used to diagnose Horner syndrome
- Describe the characteristics of physiologic anisocoria
- Describe how lesions in the afferent, central, and efferent pupillary pathways will affect the direct, consensual, and near responses, as well as normal and abnormal pupillary reactions that occur during the swinging flashlight test

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apriclonidin

respond normal to light, normal to dilatation, fluctuate troug
~~the day, usually pretty small difference between~~ two eyes

parasympathetic problem: no direct response

swinging flashlight: dilatate difference RAPD

Visual Pathway

- Describe how areas of the visual field are represented on the retina
- Describe how the visual fibers exit the eye through the optic disc and how this would affect the visual field
- Describe where the visual fibers are located in the optic nerve
- Describe how the fibers travel through the chiasm and how this affects the visual field

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gesichtsfeld vs. netzhautfeld

3 visual fields nasal step, arquid, altitudinal vf

junctional scotoma, hemi--> chiasma lesion

Visual Pathway (cont.)

- Describe the location of inferior, superior, and macular fibers in the optic tract, lateral geniculate nucleus, optic radiations, and occipital lobe.
- Describe the location of the macular and peripheral vision representation on the visual cortex and how this affects the visual field.
- Correlate lesions in areas of the visual pathway with the visual field and vice versa

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temporal fibre are more lateral

lower retina lateral side lingual

superior fibres more medialbridal lobe cunous gyros

macula is most posterior, posterior central defect
~~respects horizontal line~~

anterior lesion more peripheral, spare macular

good visual acuity

R: posterior lingual gyrus creates
left superior macular (because it's posterior)